

MACHINE LEARNING JOURNEY — DAY 1 NOTES

Artificial Intelligence, Classical Programming & Machine Learning

Beginner-Friendly Study Notes with Diagrams, Tables & Practice Questions

Topics Covered:

1. What is Artificial Intelligence (AI)?
2. What is Classical Programming?
3. What is Machine Learning?
4. Real-Life Examples of AI, Classical Programming & ML

Prepared for Coursera Machine Learning Course — Zero Prior Knowledge Assumed

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1. Classical Programming — the "old way" computers solve problems
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Classical Programming

A. Simple Definition

Classical Programming means a human writes exact step-by-step instructions (called **rules**), the computer follows those rules on some **input**, and always produces a fixed **output**. The computer never "decides" anything on its own — it only obeys.

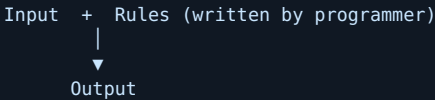
B. Why Do We Need It?

Before machines could "learn," every single behaviour of a computer had to be typed out by a programmer. This is still needed today whenever the logic of a task is 100% clear and does not change — for example, adding two numbers, checking a password, or turning a traffic light red after 30 seconds. Classical programming solves the problem of *"how do I tell a computer exactly what to do, in a way it cannot misunderstand?"*

C. Detailed Explanation

In classical programming, the programmer already knows the logic before writing any code. They think through every possible situation and write an "if this happens, do that" rule for it. The computer does not guess, estimate, or improve with experience — it simply executes the written rules, every time, in the same way.

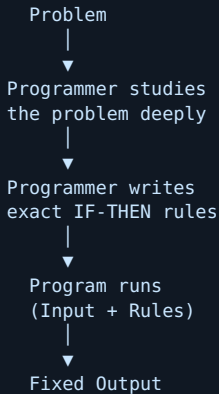
The Core Idea



D. How It Works (Step by Step)

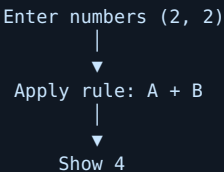
1. The programmer studies the problem and understands exactly what should happen in every case.
2. The programmer writes rules using a programming language (if/else, loops, formulas).
3. The program receives an **input** (numbers, text, button press, sensor reading).
4. The program applies the fixed rules to that input.
5. The program produces a fixed, predictable **output**.
6. If the rules were wrong, the programmer must go back and manually rewrite them — the computer cannot fix itself.

General Flow



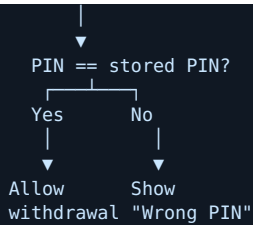
E. Real-Life Examples with Flowcharts (at least 5)

- 1. Calculator** — You press 2 + 2, the calculator follows the fixed rule of addition and always shows 4.

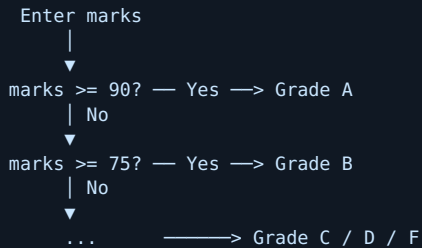


- 2. ATM Machine** — You enter your PIN; the machine checks it against a fixed rule.

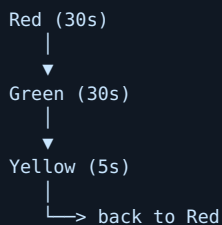




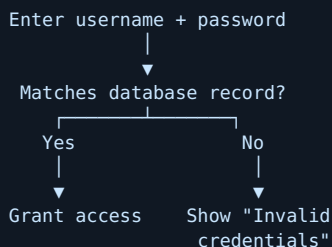
3. Student Grading System — Marks are converted to a grade using fixed cut-offs.



4. Traffic Light — A timer switches the light in a fixed, repeating order.



5. Login System — Username and password are compared with stored records.



F. Advantages

- Behaviour is 100% predictable — same input always gives the same output.
- Easy to test, debug, and explain to others.
- Fast to run because there is no learning or guessing involved.
- Works perfectly for tasks with clear, unchanging rules.

G. Limitations

- Cannot handle tasks where rules are too complex or unknown (e.g., recognising a face).
- Every new situation needs a programmer to manually add a new rule.
- Does not improve automatically with more data or experience.
- Breaks or gives wrong results when a real-world case was not anticipated by the programmer.

H. Key Points

- Formula: **Data + Rules = Output**
- Rules are written entirely by a human.
- Best for problems with clear logic (calculators, ATMs, traffic lights).
- Cannot "learn" — it only "obeys."

I. Common Interview Questions

Q1. What is classical programming?

A: An approach where a human writes fixed rules, and the computer applies those rules to input data to produce output.

Q2. Why can't classical programming handle image recognition well?

A: Because nobody can write a complete set of fixed rules covering every possible face, angle, or lighting condition — the logic is too complex to hand-write.

Q3. Give one real-world example where classical programming is still the best choice.

A: A calculator or an ATM PIN check — the rules are simple, fixed, and never need to change.

J. Quick Revision

Classical Programming in one line: Human writes the rules, computer just follows them.

Formula: Data + Rules = Output

Best for: Simple, fixed-logic problems (calculators, ATMs, login systems, traffic lights).

Artificial Intelligence (AI)

A. Simple Definition

Artificial Intelligence is the science of making computers do things that would normally need **human intelligence** — such as understanding language, recognising images, making decisions, or solving problems.

B. Why Do We Need It?

Many real-world tasks are too complex, too fast, or too repetitive for humans to handle alone — reading millions of emails to catch spam, driving a car safely every second, or translating speech in real time. AI lets computers take on these human-like tasks at massive scale and speed.

C. Detailed Explanation

AI is not one single technology — it is a broad **goal**: making machines "smart." Machine Learning (which you will study next) is simply the most popular current *method* used to achieve that goal. Think of AI as the destination, and Machine Learning as the vehicle most commonly used to get there today.

Relationship Diagram



D. How It Works



Types of AI (Brief Introduction)

Type	Meaning	Example Today
Narrow AI	Good at only ONE specific task	Siri, spam filters, face unlock (this is what exists today)
General AI	Could perform ANY intellectual task a human can	Does not exist yet — still research/theory
Super AI	Would surpass human intelligence in all areas	Purely theoretical / futuristic concept

E. Real-Life Examples (5+)

- 1. Siri / Google Assistant / Alexa** — Understand your spoken question and reply in natural language.
- 2. Self-Driving Cars** — Perceive the road, other cars, and pedestrians, then decide steering and braking.
- 3. Face Recognition** — Unlocks your phone by recognising your face pattern.
- 4. Chatbots** — Answer customer questions on websites automatically, 24/7.
- 5. Recommendation Systems** — Suggest the next video, song, or product you are likely to like.

F. Advantages

- Can work 24/7 without getting tired.

- Handles huge amounts of data far faster than humans.
- Reduces human error in repetitive tasks.
- Can find patterns humans might miss.

G. Challenges / Limitations

- Needs large amounts of good-quality data to work well.
- Can be biased if the training data is biased.
- Struggles with tasks needing genuine common sense or emotion.
- Raises questions about jobs, privacy, and ethics.

Future of AI

AI is expected to become more integrated into everyday tools — smarter assistants, more accurate medical diagnosis, and better personalised education — while researchers continue working on making it more transparent, fair, and trustworthy.

H. Key Points

- AI = the broad goal of making machines act intelligently.
- Today's AI is mostly **Narrow AI** — good at one task only.
- Machine Learning is the main technique used to build modern AI.

I. Common Interview Questions

Q1. What is the difference between Narrow AI and General AI?

A: Narrow AI is designed for one specific task (like voice recognition), while General AI (still theoretical) would be able to perform any intellectual task a human can.

Q2. Is all AI Machine Learning?

A: No. AI is the broader goal; Machine Learning is one popular method used to achieve it. Some older AI systems used only fixed rules, not learning.

Q3. Name three everyday applications of AI.

A: Voice assistants (Siri/Alexa), face unlock on phones, and recommendation systems (Netflix/YouTube).

J. Quick Revision

AI in one line: Making machines act intelligently, like a human would.

Types: Narrow AI (exists today) → General AI (theoretical) → Super AI (theoretical).

Remember: AI is the goal; Machine Learning is the method.

Machine Learning (ML)

A. Simple Definition

Machine Learning is a way of building AI where, instead of a human writing fixed rules, the computer **looks at lots of examples (data)** and **figures out the rules (patterns) by itself**. Once it has learned these patterns, it can make predictions on new, unseen data.

B. Why Do We Need It?

For many real problems — like recognising handwriting, predicting house prices, or filtering spam — nobody can write a complete, exact set of rules by hand. There are too many exceptions and variations. Machine Learning solves this by letting the computer discover the rules on its own from real examples, instead of a human guessing them.

Difference Between AI and Machine Learning

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
What it is	The broad goal of making machines smart	One method/technique to reach that goal
Scope	Includes ML, rule-based systems, robotics, etc.	A subset that specifically learns from data
Example	A chess-playing program (rule-based or ML)	A spam filter that learns from past emails

C. Detailed Explanation

Instead of the programmer supplying rules, in ML the programmer supplies **examples** (data) along with the correct answers. The learning algorithm studies these examples repeatedly and gradually works out a general pattern. This pattern is stored inside something called a **model**. Once trained, the model can look at brand-new data it has never seen and make a sensible guess (prediction).

Classical Programming vs Machine Learning — the Big Shift



D. How It Works — Step by Step



1. **Collect Data** — gather many real examples (e.g., 10,000 past emails marked spam/not spam).
2. **Training** — feed this data into a learning algorithm; it studies the patterns (e.g., which words appear often in spam).
3. **Model** — the algorithm produces a model, which is basically the "learned rules," stored as numbers.
4. **Testing** — the model is checked on new examples it has NOT seen before, to see how accurate it is.
5. **Prediction** — once accurate enough, the model is used on brand-new, real data to make decisions.

What is Data? What is a Pattern? What is a Prediction?

Term	Simple Meaning	Example
Data	Raw examples we feed to the model	1000s of past emails
Pattern	A repeated trend the model notices in the data	Emails with "Win money now" are usually spam

Prediction	The model's guess about new, unseen data	"This new email is 92% likely to be spam"
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Model, Training, Testing, Prediction — In Very Simple Words

- **Model** — think of it as the "brain" the computer builds after studying examples.
- **Training** — the "studying" phase, where the model looks at known examples again and again.
- **Testing** — a "mock exam" using fresh examples to check how well the model actually learned.
- **Prediction** — the "real exam," where the model answers questions about brand-new data.

E. Real-Life Examples (5+)

1. **Gmail Spam Filter** — Learns from millions of emails which ones are spam.
2. **Netflix/YouTube Recommendations** — Learns your watching habits to suggest new content.
3. **Weather Prediction** — Learns from years of past weather data to forecast tomorrow.
4. **Fraud Detection** — Learns normal spending patterns to flag unusual transactions.
5. **Medical Diagnosis Assistance** — Learns from thousands of medical images to help detect diseases.

F. Advantages

- Can handle problems too complex for hand-written rules.
- Improves automatically as more data becomes available.
- Can discover hidden patterns humans might never notice.
- Scales well to huge, real-world datasets.

G. Limitations

- Needs a large amount of good-quality data to learn well.
- Predictions can be wrong — it is a "best guess," not a guarantee.
- Can inherit biases present in the training data.
- Models can be hard to fully explain ("black box" behaviour).

H. Key Points

- Formula: **Data + Output (examples) = Learning → Creates Rules**
- ML flips classical programming: instead of writing rules, we provide examples and let the computer find the rules.
- Core pipeline: Data → Training → Model → Testing → Prediction.

I. Common Interview Questions

Q1. What is Machine Learning in simple words?

A: A method where a computer learns patterns from example data instead of being given fixed rules by a programmer.

Q2. What is the difference between training data and testing data?

A: Training data is used to teach the model patterns; testing data is separate, unseen data used to check how well the model actually learned.

Q3. What is a "model" in Machine Learning?

A: The output of the training process — a saved set of learned patterns that can be used to make predictions on new data.

Q4. Why does Machine Learning need a lot of data?

A: Because patterns become reliable only when the algorithm sees enough varied examples; too little data leads to poor, unreliable predictions.

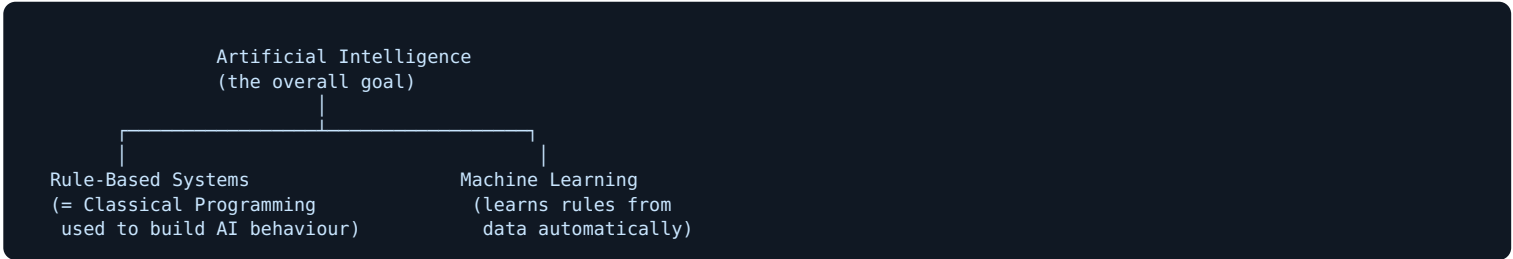
J. Quick Revision

ML in one line: Computer learns rules/patterns from data instead of being told the rules.
Formula: Data + Output examples = Learning → Creates Rules
Pipeline: Data → Training → Model → Testing → Prediction

AI vs Classical Programming vs Machine Learning

Aspect	Classical Programming	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	Human writes exact rules for the computer to follow	Broad goal of making machines act intelligently	Method where computer learns rules from data
Inputs	Data + Rules	Varies (data, rules, sensors, etc.)	Data + Correct examples (labels)
Rules	Written manually by programmer	May be written OR learned	Learned automatically from data
Learning	None — never improves on its own	Depends on the approach used	Yes — improves as more data is added
Flexibility	Low — breaks on unseen cases	Varies	High — adapts to new patterns
Examples	Calculator, ATM, traffic light	Siri, self-driving cars, chatbots	Spam filter, Netflix recommendations
Advantages	Predictable, simple, fast to build	Automates human-like tasks at scale	Handles complex, hard-to-hand-code problems
Limitations	Cannot handle unclear/complex logic	Can be biased, needs lots of resources	Needs large quality data; not always explainable
Real-Life Application	Billing systems, login pages	Voice assistants, robotics	Fraud detection, medical diagnosis support

Where They Fit Together



Memory Anchor: Classical Programming needs a human to think of every rule. Machine Learning needs a human to only collect good examples — the computer thinks of the rules itself. AI is simply the name of the goal both approaches are trying to reach.

Daily-Life Examples Explained Step by Step

1. YouTube Recommendations — YouTube records what you watch, like, and skip → the ML model finds patterns similar to other users → it predicts videos you are likely to enjoy next → shows them on your homepage.

2. Netflix Recommendations — Netflix studies your viewing history and ratings → compares your taste with millions of other viewers → the model predicts shows matching your taste → arranges your homepage rows accordingly.

3. Instagram Feed — The app tracks what posts you like, comment on, and how long you view them → the model learns your interests → it ranks and shows posts it predicts you will engage with most.

4. Google Maps — Collects live location data from many phones on the road → the model detects traffic patterns → predicts the fastest route and estimated arrival time → updates the route in real time as conditions change.

5. Gmail Spam Filter — The model was trained on millions of emails labelled spam/not-spam → it learned patterns common in spam (certain words, links, sender behaviour) → checks each new incoming email against these patterns → moves suspicious ones to the Spam folder.

6. Face Unlock — Your phone stores a mathematical "map" of your face during setup → each time you unlock, the camera captures your face → the model compares the new scan to the stored map → unlocks if it predicts a strong enough match.

7. Online Shopping Suggestions — Tracks items you view, search, and buy → the model finds patterns of "customers who bought X also bought Y" → predicts and displays products you're likely to purchase next.

8. Weather Prediction — Historical weather data (temperature, humidity, pressure, wind) is fed into a model → the model learns patterns linking these factors to outcomes → predicts tomorrow's weather based on today's readings.

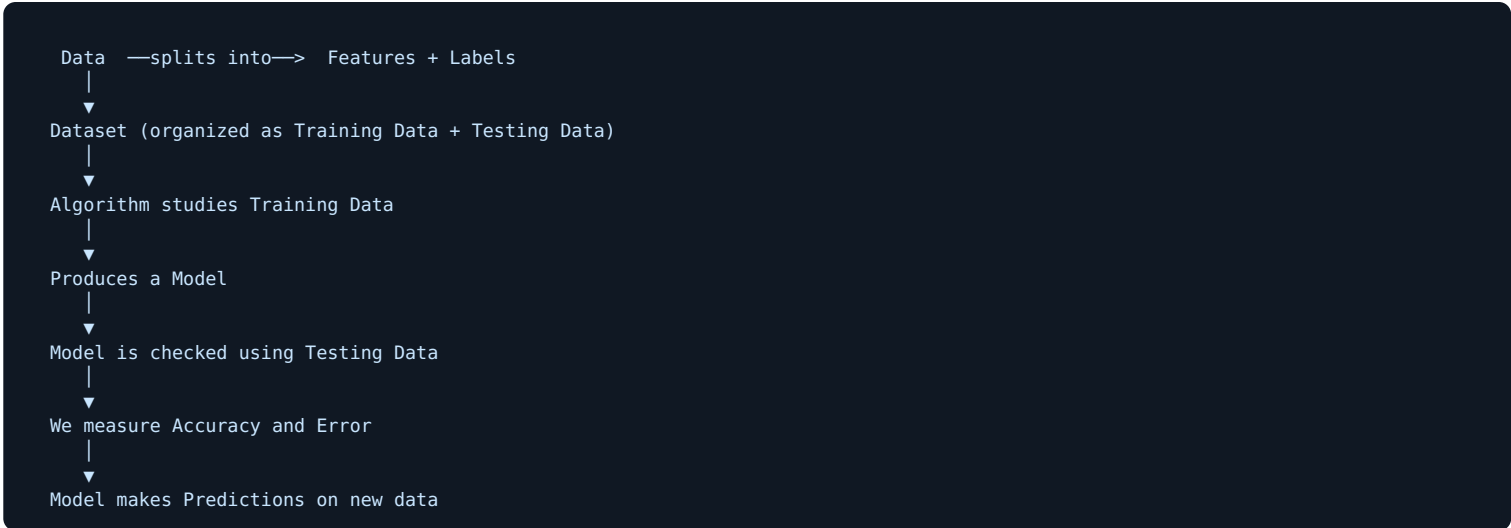
9. Medical Diagnosis Support — The model is trained on thousands of labelled medical scans (disease/no disease) → learns visual patterns linked to particular conditions → analyses a new patient's scan → predicts the likelihood of a condition, helping doctors decide faster.

10. Fraud Detection — The model learns what "normal" spending looks like for each customer → when a new transaction happens, it's compared against this pattern → if it looks unusual (e.g., a huge purchase in another country), the transaction is flagged or blocked.

11. Voice Assistants (Siri/Alexa/Google Assistant) — Your spoken words are converted to text → the model interprets the meaning/intent → matches it with an appropriate action (set timer, play music, answer a question) → responds back in speech.

Things Every Beginner Should Know

How These Terms Connect



Term	Simple Explanation	Example
Data	Raw facts or numbers collected, before they are processed into meaning	A list of house sizes and prices
Information vs Data	Data is raw; Information is data that has been processed to be useful	Data: "3, 2, 1500" → Information: "3-bed, 2-bath, 1500 sq ft house"
Dataset	A structured collection of data, usually in rows and columns	A spreadsheet of 1,000 houses with size, location, price
Features	The input details/columns used to make a prediction	Size, location, number of rooms
Labels	The correct answer/output linked to each example (used in training)	The actual price of each house
Algorithm	The step-by-step method used to find patterns in data	Linear Regression, Decision Tree
Model	What you get after an algorithm has learned from data	A trained "house price predictor"
Prediction	The model's guess/output for new, unseen input	"This house will cost about \$250,000"
Training Data	The data used to teach the model	800 out of 1,000 houses
Testing Data	Separate data used to check the model's performance	The remaining 200 houses
Accuracy	How often the model's predictions are correct	"The model is 92% accurate"
Error	The difference between the model's prediction and the real answer	Predicted \$250,000, actual was \$240,000 → error = \$10,000

Memory Tricks & Mnemonics

Classical Programming

Data + Rules = Output
(Human gives BOTH the data AND the rules)

Machine Learning

Data + Output = Learning
↓
Creates Rules
(Human gives data AND correct answers;
the computer figures out the rules itself)

Remembering the 3 Types of AI

N-G-S → "Narrow, General, Super"
N = Narrow AI = Exists NOW (one task only)
G = General AI = theoretical, "human-level"
S = Super AI = theoretical, "beyond human"

Remembering the ML Pipeline

D-T-M-T-P → "Do The Model's Testing, Please"
D = Data
T = Training
M = Model
T = Testing
P = Prediction

AI vs ML — One-Line Anchor

"AI is the **destination** (a smart machine). ML is the **vehicle** most used to get there (learning from data)."

20 Multiple Choice Questions (with Answers)

1. Classical programming requires:

- a) Data only b) Rules only c) Data + Rules d) Nothing

Answer: c) Data + Rules

2. In Machine Learning, who writes the rules?

- a) The programmer b) The computer, by learning from data c) No one d) The user

Answer: b) The computer, by learning from data

3. AI stands for:

- a) Automated Internet b) Artificial Intelligence c) Applied Information d) Advanced Interface

Answer: b) Artificial Intelligence

4. Which type of AI exists in the real world today?

- a) Narrow AI b) General AI c) Super AI d) None

Answer: a) Narrow AI

5. A calculator is an example of:

- a) Machine Learning b) Classical Programming c) General AI d) Deep Learning

Answer: b) Classical Programming

6. In ML, "training" means:

- a) Testing final results b) Teaching the model using data c) Deleting data d) Writing manual rules

Answer: b) Teaching the model using data

7. A model in Machine Learning is:

- a) A type of computer b) The learned pattern from training data c) A dataset d) A programming language

Answer: b) The learned pattern from training data

8. Testing data is used to:

- a) Train the model b) Check the model's performance on unseen data c) Delete errors d) Replace training data

Answer: b) Check the model's performance on unseen data

9. Which of these is NOT an example of AI?

- a) Siri b) Self-driving cars c) A basic calculator d) Chatbots

Answer: c) A basic calculator

10. Features in a dataset refer to:

- a) The correct answers b) The input columns used for prediction c) The algorithm d) The error rate

Answer: b) The input columns used for prediction

11. Labels in a dataset refer to:

- a) Input columns b) The correct output/answer for each example c) The algorithm name d) The testing method

Answer: b) The correct output/answer for each example

12. Accuracy measures:

- a) How fast the model runs b) How often predictions are correct c) The size of the dataset d) The programming language used

Answer: b) How often predictions are correct

13. Which formula represents classical programming?

- a) Data + Output = Learning b) Data + Rules = Output c) Model + Error = Data d) Rules + Learning = Data

Answer: b) Data + Rules = Output

14. Which formula represents Machine Learning?

- a) Data + Rules = Output b) Data + Output = Learning → Rules c) Output = Rules d) Data = Prediction

Answer: b) Data + Output = Learning → Rules

15. Gmail's spam filter is an example of:

- a) Classical Programming b) Machine Learning c) Neither d) Hardware design

Answer: b) Machine Learning

16. A traffic light controlled by a fixed timer is an example of:

- a) Machine Learning b) Classical Programming c) General AI d) Super AI

Answer: b) Classical Programming

17. Which is the correct ML pipeline order?

- a) Model → Data → Training → Prediction b) Data → Training → Model → Testing → Prediction c) Prediction → Data → Model → Testing → Data → Training

Answer: b) Data → Training → Model → Testing → Prediction

18. Error in Machine Learning refers to:

- a) A computer crash b) The difference between predicted and actual results c) A missing dataset d) A coding mistake only

Answer: b) The difference between predicted and actual results

19. Which statement is TRUE?

- a) All AI is Machine Learning b) Machine Learning is one method used to build AI c) Machine Learning and AI are unrelated d) Classical Programming is a type of Machine Learning

Answer: b) Machine Learning is one method used to build AI

20. A key limitation of Machine Learning is:

- a) It works with zero data b) It needs large, good-quality data to work well c) It never makes mistakes d) It cannot be used in real life

Answer: b) It needs large, good-quality data to work well

Short & Long Questions

10 Short Questions

1. Define Artificial Intelligence in one sentence.
2. What is the main difference between Classical Programming and Machine Learning?
3. What is a dataset?
4. Name two examples of Narrow AI.
5. What is the purpose of testing data?
6. What does "training a model" mean?
7. What is a prediction in Machine Learning?
8. Give one example of ML used in daily life.
9. What is the difference between a feature and a label?
10. Why can't classical programming handle face recognition well?

10 Long Questions

1. Explain, with a diagram, how classical programming works from input to output.
2. Describe the complete Machine Learning pipeline, from data collection to prediction.
3. Compare AI, Classical Programming, and Machine Learning using a table with at least 5 points.
4. Explain the three types of AI (Narrow, General, Super) with examples.
5. Using the example of a spam filter, explain how a Machine Learning model is trained and used.
6. What are the advantages and limitations of Machine Learning? Explain with examples.
7. Explain the terms: dataset, features, labels, training data, and testing data, using an example of predicting house prices.
8. Describe five real-life examples where Machine Learning is used, and explain briefly how each works.
9. Why is Machine Learning needed even though Classical Programming already exists? Explain with reasoning.
10. Explain the relationship between Artificial Intelligence and Machine Learning with a diagram.

Interview & Scenario-Based Questions

10 Interview Questions (Beginner Level)

1. What is the difference between AI, Machine Learning, and Deep Learning (briefly)?

A: AI is the broad goal of intelligent machines. ML is a method that learns from data. Deep Learning is a subtype of ML that uses layered neural networks.

2. Can you explain "training" and "testing" in your own words?

A: Training is when the model studies example data to learn patterns; testing is checking how well it learned using data it has not seen before.

3. Why is data important in Machine Learning?

A: The model's entire "knowledge" comes from data — poor or insufficient data leads to poor, unreliable predictions.

4. What happens if a model is tested on the same data it was trained on?

A: The accuracy score becomes misleading because the model may have just memorised the training data rather than learned a general pattern.

5. Give an example of Narrow AI you use daily.

A: A voice assistant like Siri, which is very good at understanding speech but cannot do unrelated tasks like driving a car.

6. What is the biggest limitation of classical programming compared to ML?

A: It cannot handle problems where the rules are too complex or unknown to write by hand, and it never improves automatically.

7. How would you explain "prediction" to a non-technical person?

A: It's the computer's educated guess about something new, based on patterns it noticed in past examples.

8. What is "error" in a Machine Learning model, and why does it matter?

A: It's the gap between the model's prediction and the real answer; tracking it helps us know how good or bad the model is.

9. Is more data always better for Machine Learning?

A: Generally yes, more good-quality, relevant data helps the model learn better patterns — but poor-quality or biased data can hurt performance regardless of quantity.

10. Why do you think companies like Netflix use Machine Learning instead of hand-written rules?

A: Because everyone's taste is different and constantly changing — it's impossible to hand-write personalised rules for millions of users, but ML can learn each person's pattern automatically.

5 Scenario-Based Questions

1. A bank wants to detect fraudulent transactions instantly. Would you use Classical Programming or Machine Learning, and why?

A: Machine Learning — fraud patterns are complex, constantly evolving, and cannot be fully captured by fixed hand-written rules.

2. A vending machine needs to give change based on the amount inserted. Which approach fits best, and why?

A: Classical Programming — the logic (amount inserted minus item price) is simple, fixed, and doesn't need to "learn" anything.

3. A hospital wants to predict which patients are at risk of a disease based on thousands of past patient records. Which approach fits, and why?

A: Machine Learning — the relationships between symptoms and disease risk are too complex for hand-written rules but can be learned from historical patient data.

4. An app wants to automatically translate voice messages into text in real time. Which approach is more suitable, and why?

A: Machine Learning — human speech has too many accents, tones, and variations to cover with fixed rules; the model learns these patterns from vast amounts of speech data.

5. A simple website login page needs to check if a username and password match a database. Which approach is best, and why?

A: Classical Programming — this is a simple fixed comparison rule (match or no match), with no need for learning from data.

One-Page Quick Revision Sheet

Core Definitions

- **Classical Programming** — Human writes fixed rules; computer follows them exactly.
- **Artificial Intelligence (AI)** — The broad goal of making machines act intelligently, like humans.
- **Machine Learning (ML)** — Computer learns rules/patterns automatically from data (a method used to build AI).

Formulas to Remember

Classical Programming: Data + Rules = Output

Machine Learning: Data + Output = Learning → Creates Rules

Key Diagram



Comparison At a Glance

Feature	Classical Programming	Machine Learning
Rules	Written by programmer	Learned from data
Data	Used as input	Used for learning
Output	Fixed	Predictive
Flexibility	Low	High

ML Pipeline

Data → Training → Model → Testing → Prediction

Types of AI

Narrow AI (exists today, one task) → **General AI** (theoretical, human-level) → **Super AI** (theoretical, beyond human)

Common Mistakes to Avoid

- Do not say "AI and ML are the same thing" — ML is a subset/method used to build AI.
- Do not confuse "training data" with "testing data" — they must be kept separate for a fair accuracy check.
- Do not assume Classical Programming is "outdated" — it is still the right tool for simple, fixed-logic problems.
- Do not forget: a model's prediction is a probability/best guess, not a guaranteed fact.

Golden One-Liner

"Classical Programming: I give the rules. Machine Learning: I give the examples, the computer finds the rules. AI: the big goal both are working toward."